

UniScout / College Search Platform — Agent Context

Purpose: Single source of truth for AI agents continuing development.

Product: Indian college discovery, search, filter, compare, and (planned) save.

Brand in UI: UniScout (`src/components/navbar.tsx`)

1. Quick facts

Item	Value
Repo folder	<code>collegesearchplatform`</code>
Package name	<code>collegesearchplatform`</code>
Framework	Next.js 16.2.6 (App Router, Turbopack in dev)
React	19.2.4
Language	TypeScript 5.9.3 (strict)
Styling	Tailwind CSS v4 , shadcn/ui (<code>base-lyra`</code> style)
Database	PostgreSQL via Prisma 7.8.0 (Neon-compatible)
Auth (dependency only)	<code>@clerk/nextjs`</code> 7.4.0 — not wired yet
Path alias	<code>@/*`</code> <code>!`</code> <code>src/*`</code>

2. Runtime & toolchain versions

Verified on Windows (May 2026):

Tool	Version
Node.js	v22.22.0
npm	11.15.0
Prisma CLI	7.8.0
@prisma/client	7.8.0
Prisma Studio	0.27.3

Install deps: `npm install`

Dev server: `npm run dev`` `http://localhost:3000`

Production build: `npm run build`` then `npm run start`

Lint: `npm run lint`

3. Environment variables

Variable	Required	Used by
<code>DATABASE_URL`</code>	Yes	(Prisma, seed) <code>prisma.config.ts`</code> , <code>src/lib/prisma.ts`</code>

Planned (not in codebase yet):

Variable	For
<code>NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY`</code>	Clerk auth
<code>CLERK_SECRET_KEY`</code>	Clerk server
<code>CLERK_WEBHOOK_SECRET`</code>	User sync webhooks

Never commit `.env`. Agents must not print secret values from `.env`.

4. NPM scripts

Script	Command	Notes
<code>dev`</code>	<code>next dev`</code>	Hot reload, Turbopack
<code>build`</code>	<code>next build`</code>	Typecheck + static generation

```
`start` `next start` After build
`lint` `eslint` Next ESLint config
```

Prisma (via CLI, not in package.json scripts):

```
npx prisma generate # Regenerate client after schema change
npx prisma db push # Sync schema to DB (no migration files yet)
npx prisma db seed # Runs: npx ts-node ./prisma/seed.ts
npx prisma studio # DB GUI
npx prisma migrate dev # When migrations are introduced
```

Seed config: prisma.config.ts ! migrations.seed

5. Routes & pages (App Router)

There are no REST API routes under `src/app/api/` yet. All behavior is pages + client state.

Route	File	Render	Description
-------	------	--------	-------------

<code>/</code>	<code>src/app/page.tsx</code>	Client	Main discovery: search, filters, grid, compare selection, pagination
<code>/colleges</code>	<code>src/app/colleges/page.tsx</code>	Server	Redirects to <code>/</code>
<code>/colleges/[id]</code>	<code>src/app/colleges/[id]/page.tsx</code>	Server	Detail page; <code>id</code> matches <code>**static**</code> <code>colleges-data</code> ids (<code>"1"</code> – <code>"6"</code>)
<code>/colleges/compare</code>	<code>src/app/colleges/compare/page.tsx</code>	Server	Placeholder; compare UI on home not linked here yet
<code>/saved</code>	<code>src/app/saved/page.tsx</code>	Server	Placeholder; saved colleges need Clerk + Prisma

Build output (static/dynamic):

- `%E` /, /colleges, /colleges/compare, /saved``
- `f` /colleges/[id]`` (dynamic)

6. Recommended API endpoints (to implement)

Agents should add these under `src/app/api/` when connecting Prisma to the UI.

Colleges

Method	Path	Purpose
--------	------	---------

<code>GET</code>	<code>/api/colleges</code>	List with query: <code>q</code> , <code>type</code> , <code>minFees</code> , <code>maxFees</code> , <code>location</code> , <code>sort</code> , <code>page</code> , <code>limit</code>
<code>GET</code>	<code>/api/colleges/[id]</code>	Single college by DB <code>id</code> (cuid)
<code>POST</code>	<code>/api/colleges</code>	Admin create (auth + role)
<code>PATCH</code>	<code>/api/colleges/[id]</code>	Admin update
<code>DELETE</code>	<code>/api/colleges/[id]</code>	Admin delete

Saved colleges (requires auth)

Method	Path	Purpose
--------	------	---------

<code>GET</code>	<code>/api/saved</code>	Current user's saved colleges
<code>POST</code>	<code>/api/saved</code>	Body: <code>{ collegeld }</code>
<code>DELETE</code>	<code>/api/saved/[collegeld]</code>	Unsave

Users (Clerk webhook / sync)

Method	Path	Purpose
--------	------	---------

<code>POST</code>	<code>/api/webhooks/clerk</code>	Sync <code>User</code> row on sign-up
-------------------	----------------------------------	---------------------------------------

Compare (optional)

Method	Path	Purpose
--------	------	---------

<code>GET</code>	<code>/api/compare?ids=id1,id2</code>	Side-by-side payload for 2–4 ids
------------------	---------------------------------------	----------------------------------

Use `src/lib/prisma.ts` in route handlers. Return JSON; map DB fields to frontend College shape (see §8).

7. Database schema (Prisma)

File: prisma/schema.prisma

Provider: postgresql

Migrations folder: prisma/migrations (empty — schema synced via db push)

`College`

Field	Type	Notes
-------	------	-------

`id`	`String`	@id @default(cuid()) **Not** the same as static UI ids <code>"1"–"6"</code>
`name`	`String`	
`location`	`String`	e.g. <code>"New Delhi, DL"</code> — state code used for region filters
`fees`	`Int`	Annual fees in INR
`rating`	`Float`	e.g. 4.8
`placementPercentage`	`Int`	0–100
`description`	`String`	
`image`	`String`	URL
`courses`	`String[]`	Maps to UI <code>`programs`</code>
`createdAt`	`DateTime`	

Not in DB yet: type, ranking, acceptanceRate, enrollment — only in static colleges-data.ts. Extend schema if filters should be DB-backed.

`User`

Field	Type	Notes
-------	------	-------

`id`	`String`	@id @default(cuid()) Internal id
`clerkId`	`String`	@unique Clerk user id
`email`	`String`	@unique
`createdAt`	`DateTime`	
`savedColleges`	`SavedCollege[]`	

`SavedCollege`

Field	Type	Notes
-------	------	-------

`id`	`String`	@id @default(cuid())
`userId`	`String`	FK !' `User.id`
`collegeId`	`String`	**No FK relation to College in schema** — store id string only
`createdAt`	`DateTime`	

Gap: Add college College @relation(...) on SavedCollege when wiring saves.

Seed

- File: ``prisma/seed.ts``
- Imports: ``src/lib/prisma.ts``
- Deletes all colleges, inserts 6 (IIT Delhi, DTU, NSUT, IIT Bombay, BITS Pilani, VIT Vellore)
- Run: ``npx prisma db seed``

8. Data model: UI vs database

Frontend type (``src/components/college-card.tsx``)

```
interface College {  
  id: string  
  name: string  
  location: string  
  type: string           // "public" | "private" | ...  
  ranking: number  
  acceptanceRate: number  
  tuition: number        // INR
```

```

enrollment: number
image: string
featured?: boolean
programs: string[]
}

```

Static data (`src/lib/colleges-data.ts`)

- ****6 colleges****, ids ``"1" ... `6``
- Powers: home page, detail page, compare bar
- Fields include ``type``, ``ranking``, ``acceptanceRate``, ``tuition``, ``programs``

Prisma !' UI mapper (implement in `src/lib/college-mapper.ts`)

UI field DB field / rule

```

`id`    `id`
`name`    `name`
`location`    `location`
`tuition`    `fees`
`programs`    `courses`
`rating`    `rating` (UI also uses for display; card shows acceptance/enrollment from static only today)
`type`, `ranking`, `acceptanceRate`, `enrollment`    Defaults or new columns

```

Critical: Until API integration, `/colleges/[id]` only works for static ids, not DB cuids.

9. Home page behavior (`src/app/page.tsx`)

Client component. State:

```

type Filters = {
  type: string[]                    // "public", "private", etc.
  tuitionRange: [number, number]    // default [0, 500000]
  acceptanceRate: string            // "all" | "under-10" | "10-25" | "25-50" | "over-50"
  location: string                  // "all" | "north" | "south" | "west" | "east" | "central"
  ranking: string                   // "all" | "top-25" | "top-50" | "top-100"
}

```

Constants: `src/lib/constants.ts` — `TUITION_MIN`, `TUITION_MAX` (500000), `TUITION_STEP` (10000)

Location region !' state codes (India)

Filter State codes in ``location`` string

```

`north`    DL, HR, PB, UP, UK, JK, HP, CH
`south`    TN, KA, KL, AP, TS, PY
`west`    MH, RJ, GJ, GA
`east`    WB, OR, BH, JH, AS, SK, ML, MN, TR, NL
`central`    MP, CG

```

Sort options

ranking (default), name, tuition-low, tuition-high, acceptance

Pagination

`ITEMS_PER_PAGE` = 9 from `colleges-data.ts`

Compare

- Max ****4**** colleges selected
- ``CompareBar`` fixed bottom; "Compare Now" disabled until 2+ selected
- Does not navigate to ``/colleges/compare`` yet

10. Shared libraries

File Role

```
`src/lib/prisma.ts`    Singleton PrismaClient + `PrismaPg` adapter (Prisma 7)
`src/lib/colleges-data.ts`    Static college list + `ITEMS_PER_PAGE`
`src/lib/constants.ts`    Tuition filter bounds
`src/lib/format.ts`    `formatNumber`, `formatCurrency` — **always use for INR display** (`en-US` locale prevents hydration bugs)
`src/lib/utils.ts`    `cn()` — clsx + tailwind-merge
```

11. Components map

Component Path Role

```
`Navbar`    `src/components/navbar.tsx`    Logo, Browse `/`, Compare `/colleges/compare`
`SearchBar`    `src/components/search-bar.tsx`    Text search (name, location, programs)
`FiltersSidebar`    `src/components/filters-sidebar.tsx`    Desktop filters
`MobileFilters`    `src/components/mobile-filters.tsx`    Sheet wrapper for filters
`CollegeCard`    `src/components/college-card.tsx`    Card + compare checkbox + link to detail
`CompareBar`    `src/components/compare-bar.tsx`    Sticky compare tray
`PaginationControls`    `src/components/pagination-controls.tsx`    Page navigation
```

UI kit (shadcn) — ****actively typechecked****

button, badge, card, checkbox, dialog, input, label, select, sheet, slider

Many other files under `src/components/ui/` exist but are excluded in `tsconfig.json` to avoid build noise. Safe to use; may need deps or `tsconfig` include when adopted.

12. Styling & config

File Purpose

```
`src/app/globals.css`    Tailwind v4 + CSS variables
`components.json`    shadcn config; aliases `@/components`, `@/lib`, `@/hooks`
`postcss.config.mjs`    Tailwind PostCSS
`next.config.ts`    Default (empty options)
```

Hydration rule: Never use `bare number.toLocaleString()` in SSR/client shared UI. Use `formatNumber` / `formatCurrency` from `src/lib/format.ts`.

Currency: INR (₹). Tuition/filter amounts are in rupees (e.g. 250000 = ₹2.5L).

13. TypeScript notes

- `tsconfig.json`` excludes unused shadcn UI files (see exclude list in repo)
- `ts-node`` CommonJS override for Prisma seed only
- Strict mode enabled

14. Key dependencies (pinned in package.json)

Core

Package Version

```
next    16.2.6
react / react-dom    19.2.4
typescript    ^5.9.3
tailwindcss    ^4
@tailwindcss/postcss    ^4
```

Database

```
Package  Version
prisma   ^7.8.0
@prisma/client  ^7.8.0
@prisma/adapters-pg  ^7.8.0
pg       ^8.21.0
dotenv   ^17.4.2
```

UI / forms (installed; partial use)

```
Package  Version
@radix-ui/react-*  various ^1.x-^2.x
class-variance-authority  ^0.7.1
clsx      ^2.1.1
tailwind-merge  ^3.6.0
lucide-react  ^1.16.0
@phosphor-icons/react  ^2.1.10
shadcn    ^4.8.0
next-themes  ^0.4.6
react-hook-form  ^7.76.0
zod       ^4.4.3
zustand    ^5.0.13
axios     ^1.16.1
react-hot-toast  ^2.6.0
```

Auth (not integrated)

```
Package  Version
@clerk/nextjs  ^7.4.0
```

Dev

```
Package  Version
eslint    ^9
eslint-config-next  16.2.6
ts-node   ^10.9.2
@types/node  ^20.19.41
@types/pg   ^8.20.0
```

15. Project structure

```
collegesearchplatform/
% % % prisma/
% % % % schema.prisma
% % % % seed.ts
% % % prisma.config.ts
% % % src/
% % % % app/
% % % % % layout.tsx
% % % % % page.tsx      # Home / discovery
% % % % % globals.css
% % % % % colleges/
% % % % % % page.tsx    # !' redirect /
% % % % % % [id]/page.tsx
% % % % % % compare/page.tsx
% % % % % % saved/page.tsx
% % % % % components/
```

```

% % % % navbar.tsx
% % % % college-card.tsx
% % % % filters-sidebar.tsx
% % % % search-bar.tsx
% % % % compare-bar.tsx
% % % % mobile-filters.tsx
% % % % pagination-controls.tsx
% % % % ui/          # shadcn components
% % % % lib/
% % % % prisma.ts
% % % % colleges-data.ts
% % % % constants.ts
% % % % format.ts
% % % % utils.ts
% % % % docs/
% % % % PROJECT_CONTEXT.md    # This file
% % % % project-changes-summary.md
% % % % scripts/
% % % % generate-changes-pdf.mjs
% % % % .env                # DATABASE_URL (gitignored)
% % % % package.json
% % % % tsconfig.json
% % % % next.config.ts

```

16. Development priorities (suggested)

1. **API + Prisma** — `GET /api/colleges`, replace static `collegesData` on home/detail
2. **Schema alignment** — Add `type`, `ranking`, `acceptanceRate`, `enrollment` to `College` OR drop from filters
3. **Clerk** — `middleware.ts`, sign-in, sync `User`, implement `/saved`
4. **Compare flow** — Pass selected ids to `/colleges/compare` (query or session)
5. **SavedCollege FK** — Relation to `College`, unique `[userId, collegeId]`
6. **Migrations** — Move from `db push` to `prisma migrate` for production
7. **Metadata** — Update `layout.tsx` title/description from "Create Next App" to UniScout

17. Known issues & constraints

- Home and detail pages use **static data**, not PostgreSQL
- Static ids ("1") " Prisma cuids — detail URLs break after DB switch unless mapper/slug added
- `@clerk/nextjs` installed but no `middleware.ts` or ClerkProvider
- `SavedCollege.collegeld` has no Prisma relation to `College`
- Compare page is a stub; CompareBar does not link to it
- Rankings nav link still `#` in navbar
- Importing `src/lib/prisma.ts` throws if `DATABASE\_URL` missing (intentional)

18. Agent workflow checklist

When making changes:

1. Read this file + `prisma/schema.prisma` + affected page/component

2. Prefer `formatCurrency` for any fee display
3. Keep tuition bounds in sync: `constants.ts` + `filters-sidebar` + `page.tsx`
4. After schema change: `npx prisma generate` ! `npx prisma db push` ! update seed if needed
5. Run `npm run build` before considering task done
6. Do not commit `.env` or secrets
7. Minimize scope — match existing patterns (`"use client"` on interactive pages)

19. Related docs

Document	Path
----------	------

Change log (May 2026 fixes)	<code>docs/project-changes-summary.md</code>
-----------------------------	--

PDF change summary	<code>docs/project-changes-summary.pdf</code>
--------------------	---

Default Next README	<code>README.md</code> (mostly boilerplate)
---------------------	---

Last updated: May 22, 2026